

1. (True/False) Machine Learning is a subset of Artificial Intelligence

☐ False

☒ True

✔ Nice work

Correct!

2. (True/False) Deep Learning is a subset of Machine Learning

☐ False

☒ True

✔ Nice work

Correct!

3. (True/False) Machine Learning consists in programming computers to learn from real-time human interactions

☒ False

☐ True

1. (True/False) AI Winters happened mostly due to the lack of understanding behind the theory of neural networks

1 point

☐ True

☒ False
2. Most modern applications that use computer vision, use models that were trained using this discipline:

1 point

☐ Machine Learning

☐ Artificial Intelligence

☒ Deep Learning
3. In the Machine Learning Workflow, the main goal of the Data Exploration and Preprocessing step is to:

1 point

☐ Identify what data that is best suited to find a solution to your business problem

☒ Determine how to clean your data such that you can use it to train a model

1. What is the goal of supervised learning?

1 / 1 point

- ☐ Find the target.
- ☒ Predict the labels.
- ☐ Find an underlying structure of the dataset without any labels.
- ☐ Predict the features.

✔ Nice work

The goal for supervised learning is to be able to predict the label.

2. What is deep learning?

1 / 1 point

- ☒ Deep learning is machine learning that involves deep neural networks.
- ☐ Deep learning is another name for artificial intelligence.
- ☐ Deep learning includes artificial intelligence and machine learning.
- ☐ None of the above are correct.

✔ Nice work

Deep learning is machine learning that involves using very complicated models called deep neural networks. Deep learning is a subset of machine learning.

3. When is a standard machine learning algorithm usually a better choice than using deep learning to get the job done?

- ☒ When working with small data sets.
- ☐ When the data is steady over time.
- ☐ When working with large data sets.
- ☐ None of the above are correct.

✓ Nice work

A standard machine learning algorithm is a better choice when you are working with smaller datasets, and if the data is changing a lot over time and you don't have a steady dataset.

4. What is a Turing test?

- ☐ It tests and cleans the dataset.
- ☒ It tests a machine's ability to exhibit intelligent behavior.
- ☐ It tests the dataset.
- ☐ It tests images.

✓ Nice work

In 1950, Alan Turing developed the Turing test to test a machine's ability to exhibit intelligent behavior. Alan Turing's test has served as a foundational threshold for artificial intelligence.

5. What are some of the different milestones in deep learning history?

- ☐ Deep Blue defeats a world champion chess player and TensorFlow is released
- ☒ Geoffrey Hinton's work, AlexNet, and TensorFlow
- ☐ Deep Blue defeats a world champion chess player, and AlexNet is created.
- ☐ Deep Blue defeats a world champion chess player, and Keras is released.

✔ Nice work

In 2006, the previous limitations of deep learning, namely exploding and vanishing gradients were overcome with algorithmic advancements such as Geoffrey Hinton's work on unsupervised pre-training. Neural networks are rebranded as deep learning, as we are able to train much deeper networks, networks with more layers; In 2012, a deep learning model using convolutional neural nets called AlexNet achieved a top five error of 15.3 percent; In 2015, one of the most popular libraries, TensorFlow, was built for deep learning, making it more powerful and accessible.

6. What is artificial intelligence?

- ☐ A subset of deep learning.
- ☒ Any program that can sense, reason, act, and adapt.
- ☐ A subset of machine learning
- ☐ None of the above.

✔ Nice work

Artificial intelligence is any program that can sense, reason, act, and adapt. It is essentially a machine taking any form of intelligent behavior.

7. What are two spaces within AI that are going through drastic growth and innovation?

- ☐ Deep learning and machine learning.
- ☒ Computer vision and natural language processing.
- ☐ Language processing and deep learning.
- ☐ Computer vision and deep learning.

✔ Nice work

In two spaces we are seeing drastic growth and innovation, computer vision and natural language processing. The sharp advancements in computer vision are impacting multiple areas. For example, cars getting to the point where they can drive themselves. Similarly, natural language processing is booming with vast improvements in ability to translate, determine sentiment, clustering of new articles, writing papers, and many others.

8. Why did AI flourish so much in the last years?

- ☐ Access to hardware for cleaning data
- ☐ Stylish designed computers
- ☐ Data storage in the cloud is much more expensive
- ☒ Faster and inexpensive computers and data storage

✔ Nice work

AI flourished in the last years with the cloud infrastructure now in place to store large amounts of data for much cheaper, and the plethora of new ways to capture data are now able to build larger, more

9. How does Alexa use artificial intelligence?

- ☐ Recognizes faces and pictures.
- ☒ Recognizes our voice and answers questions.
- ☐ Suggests who a person on a photo is.
- ☐ None of the above answers are correct.

✔ Nice work

Alexa, in our homes, recognizes our voice and answers questions or does tasks for us through natural language processing.

10. What are the first two steps of a typical machine learning workflow?

- ☐ Problem statement and data cleaning.
- ☒ Problem statement and data collection.
- ☐ Data collection and data transformation.
- ☐ None of the above answers is correct.

✔ Nice work

The first step of a typical machine learning workflow is the problem statement. What problem are you trying to solve? The second step is data collection. What data do you need to solve the problem?

1. Which statement about the Pandas `read_csv` function is TRUE?

- ☐ It can only read comma-delimited data.
- ☐ It reads data into a 2-dimensional NumPy array.
- ☒ It can read both tab-delimited and space-delimited data.
- ☐ It allows only one argument: the name of the file.

✔ Nice work

Correct!

2. Which of the following is a reason to use JavaScript Object Notation (JSON) files for storing data?

- ☐ Because the data is stored in a matrix format.
- ☐ Because they can store NA values.
- ☐ Because they can store NULL values.
- ☒ Because they are cross-platform compatible.

✔ Nice work

Correct!

3. The data below appears in 'data.txt', and Pandas has been imported. Which Python command will read it correctly into a Pandas DataFrame?

63.03 22.55 39.61 40.48 98.67 -0.25 AB

39.06 10.06 25.02 29 114.41 4.56 AB

68.83 22.22 50.09 46.61 105.99 -3.53 AB

- ☐ pandas.read_csv('data.txt')
- ☒ pandas.read_csv('data.txt', header=None, sep=' ')
- ☐ pandas.read_csv('data.txt', delim_whitespace=True)
- ☐ pandas.read_csv('data.txt', header=0, delim_whitespace=True)

✔ Nice work

Correct!

1. (True/False) Outliers must be very extreme to noticeably impact the fit of a statistical model.

- ☐ True
- ☒ False

✔ Nice work

Correct!

2. (True/False) Outliers should always be replaced, since they never contain useful information about the data.

- ☐ True
- ☒ False

✔ Nice work

Correct!

3. Which residual-based approach to identifying outliers compares running a model with all data to running the same model, but dropping a single observation?

- ☐ Standardized residuals
- ☐ Unstandardized residuals
- ☒ Externally-studentized residuals
- ☐ Abnormally-studentized residuals

✔ Nice work

1. What is a CSV file?

1 / 1 point

- ☐ CSV is a method of JavaScript Object Notation.
- ☐ CSV files are a standard way to store data across platforms.
- ☐ CSV makes data readily available for analytics, dashboards, and reports.
- ☒ CSV files are rows of data or values separated by commas.

✔ Nice work

Correct. CSV, or Comma Separated Value, files are rows of data or values separated by commas.

2. What are residuals?

1 / 1 point

- ☐ Residuals are a method to standardize data.
- ☐ Residuals are a method for handling identified outliers.
- ☒ Residuals are the difference between the actual values and the values predicted by a given model.
- ☐ Residuals are data removed from the dataframe.

✔ Nice work

Correct. Residuals are model prediction errors.

3. If removal of rows or columns of data is not an option, why must we ensure that information is assigned for missing data?

1 / 1 point

- ☒ Most models will not accept blank values in our data.
- ☐ Information must be assigned to prevent outliers.
- ☐ Missing data may bias the dataset.
- ☐ Assigning information for missing data improves the accuracy of the dataset.

✔ Nice work

Correct. Information must be given for every feature and label in a dataset.

4. What are the two main data problems companies face when getting started with artificial intelligence/machine learning?

1 / 1 point

- ☐ Lack of training and expertise
- ☒ Lack of relevant data and bad data
- ☐ Data sampling and categorization
- ☐ Outliers and duplicated data

✔ Nice work

Correct. Companies need to collect and organize their data to make it ready before leveraging it for machine learning.

5. What does SQL stand for and what does it represent?

1 / 1 point

- ☒ SQL stands for Structured Query Language, and it represents a set of relational databases with fixed schemas.
- ☐ SQL stands for Sequential Query Language, and it represents a set of relational databases with fixed schemas.
- ☐ SQL stands for Structured Query Language, and it represents databases that are not relational, they vary in structure.
- ☐ SQL stands for Sequential Query Language, and it represents a set of sequential databases with fixed schemas.

✔ Nice work

Correct. SQL is the set of highly structured relational databases with fixed schema.

6. What does NoSQL stand for and what does it represent?

1 / 1 point

- ☐ NoSQL stands for Not-only SQL, and it represents a set of databases that are relational, therefore, they have fixed structure.
- ☒ NoSQL stands for Not-only SQL, and it represents a set of databases that are not relational, therefore, they vary in structure.
- ☐ NoSQL stands for Non-Structured Query Language, and it represents a set of non-relational databases

7. What is a JSON file?

1 / 1 point

- ☐ JSON stands for JavaScript Object Notation, and it is a non-standard way to store the data across platforms.
- ☐ JSON stands for JavaString Object Notation, and they have very similar structure to Python Dictionaries.
- ☒ JSON stands for JavaScript Object Notation, and it is a standard way to store the data across platforms.
- ☐ JSON stands for JavaString Object Notation, and it is a standard way to store the data across platforms.

✔ Nice work

Correct. JSON stands for JavaScript Object Notation, and those files are going to be a standard way to store data across platforms.

8. What is meant by the Messy Data?

1 / 1 point

- ☐ Duplicated or unnecessary data.
- ☐ Inconsistent text and typos.
- ☐ Missing data.
- ☒ All of the above.

✔ Nice work

Correct. Duplicated or unnecessary data, inconsistent text and typos, and missing data are all examples of the messy data.

9. What is an outlier?

1 / 1 point

- ☐ Outlier is a data point that does not belong in our dataset.
- ☐ Outlier is a data point that is very close to the mean value of all observations.
- ☒ Outlier is an observation in dataset that is distant from most other observations.
- ☐ Outlier is a data point that has the highest or lowest value in the dataset.

✔ Nice work

10. How do we identify outliers in our dataset?

- ☐ We can only identify outliers by using some statistical calculations.
- ☒ We can identify outliers both visually and with statistical calculations.
- ☐ We can identify outliers only by calculating the minimum and maximum values in the dataset.
- ☐ We can only identify outliers visually through building plots.

✔ Nice work

Correct. We can use plots, such as histograms, density, and box plots, as well as making some statistical calculations, such as calculating the interquartile ranges.

1. From the options listed below, select the option that is NOT a valid exploratory data approach to visually confirm whether your data is ready for modeling or if it needs further cleaning or data processing:

- ☐ Create a panel plot that shows distributions for the dependent variable and scatter plots for all independent variables
- ☒ Train a model and identify the observations with the largest residuals
- ☐ Create visualizations for scatter plots, histograms, box plots, and hexbin plots
- ☐ Create a correlation heatmap to confirm the sign and magnitude of correlation across your features.

✓ Nice work

Correct!

2. These are two of the most common variables for data visualization:

- ☒ matplotlib and seaborn
- ☐ scipy and seaborn
- ☐ numpy and matplotlib
- ☐ scipy and numpy

✓ Nice work

Correct!

3. (True/False) You can use the pandas library to use plots.

- ☒ True
- ☐ False

✓ Nice work

1. (True/False) Classification models require that input features be scaled.

☐ True

☒ False

✓ Nice work

Correct!

2. (True/False) Feature scaling allows better interpretation of distance-based approaches.

☒ True

☐ False

✓ Nice work

Correct!

3. (True/False) Feature scaling reduces distortions caused by variables with different scales.

☒ True

1. Which scaling approach converts features to standard normal variables?

- ☐ MinMax scaling
- ☒ Standard scaling
- ☐ Nearest neighbor scaling
- ☐ Robust scaling

✔ Nice work

Correct. Standard scaling converts variables to standard normal variables.

2. Which variable transformation should you use for ordinal data?

- ☐ Standard scaling
- ☒ Ordinal encoding
- ☐ One-hot encoding
- ☐ Min-max scaling

✔ Nice work

Correct. Use ordinal encoding if there is some order to the categorical features.

3. What are polynomial features?

- ☒ They are higher order relationships in the data.
- ☐ They are represented by linear relationships in the data.
- ☐ They are lower order relationships in the data.
- ☐ They are logistic regression coefficients.

✔ Nice work

Correct. Polynomial features are estimated by higher order polynomials in a linear model, like squared, cubed, etc.

4. What does Boxcox transformation do?

- ☒ It transforms the data distribution into more symmetrical bell curve
- ☐ It makes the data more right skewed.
- ☐ It transforms categorical variables into numerical variables.
- ☐ It makes the data more left skewed

✔ Nice work

Correct. Boxcox is one of the ways we can transform our skewed dataset to be more normally distributed.

5. Select three important reasons why EDA is useful.

- ☐ To utilize summary statistics, to create visualizations, and to identify outliers
- ☒ To determine if the data makes sense, to determine whether further data cleaning is needed, and to help identify patterns and trends in the data
- ☐ To analyze data sets, to determine the main characteristics of data sets, and to use sampling to examine data
- ☐ To examine correlations, to sample from dataframes, and to train models on random samples of data

✔ Nice work

Correct. EDA helps us analyze data to summarize its main characteristics.

6. What assumption does the linear regression model make about data?

- ☐ This model assumes a transformation of each parameter to a linear relationship.
- ☒ This model assumes a linear relationship between predictor variables and outcome variables.
- ☐ This model assumes an addition of each one of the model parameters multiplied by a coefficient.
- ☐ This model assumes that raw data in data sets is on the same scale.

✔ Nice work

Correct. The linear regression model assumes a linear relationship between predictor and outcome variables.

7. What is skewed data?

- ☒ Data that is distorted away from normal distribution; may be positively or negatively skewed.
- ☐ Data that has a normal distribution.
- ☐ Raw data that has undergone log transformation.
- ☐ Raw data that may not have a linear relationship.

✔ Nice work

Correct. Often raw data, both the features and the outcome variable, can be negatively or positively skewed.

8. Select the two primary types of categorical feature encoding.

- ☒ One-hot encoding and ordinal encoding
- ☐ Log and polynomial transformation
- ☐ Encoding and scaling
- ☐ Frequency encoding and label encoding

✔ Nice work

Correct. Encoding that transforms non-numeric values to numeric values is often applied to categorical features.

9. Which scaling approach puts values between zero and one?

- ☐ Standard scaling
- ☐ Robust scaling
- ☐ Nearest neighbor scaling
- ☒ Min-max scaling

✔ Nice work

Correct. Min-max scaling converts variables to continuous variables in the (0, 1) interval by mapping minimum values to 0 and maximum values to 1.

10. Which variable transformation should you use for nominal data with multiple different values within the feature?

- ☒ One-hot encoding
- ☐ Standard scaling
- ☐ Ordinal encoding
- ☐ Min-max scaling

✓ Nice work

Correct. Use one-hot encoding if there are multiple different values within a feature.

1. (True/False) In general, the population parameters are unknown.

☒ True.

☐ False.

✔ Nice work

Correct!

2. (True/False) Parametric models have **finite** number of parameters.

☒ True.

☐ False.

✔ Nice work

Correct!

3. The most common way of estimating parameters in a parametric model is:

☒ using the maximum likelihood estimation

☐ using the central limit theorem

☐ extrapolating a non-parametric model

☐ extrapolating Bayesian statistics

✔ Nice work

Correct!

1. A p-value is:

- ☒ the smallest significance level at which the null hypothesis would be rejected
- ☐ the probability of the null hypothesis being true
- ☐ the probability of the null hypothesis being false
- ☐ the smallest significance level at which the null hypothesis is accepted

✔ Nice work

Correct!

2. Type 1 Error 1 is defined as:

- ☒ Saying the null hypothesis is false, when it is actually true
- ☐ Saying the null hypothesis is true, when it is actually false

✔ Nice work

Correct!

3. You find through a graph that there is a strong correlation between Net Promoter Score and the visual time that customers spend on a website. Select the TRUE assertion:

- ☒ There is an underlying factor that explains this correlation, but manipulating the time that customers spend on a website may not affect the Net Promoter Score they will give to the company
- ☐ To boost the Net Promoter Score of a business, you need to increase the time that customers spend on a website.

✔ Nice work

Correct!

1. Which one of the following is common to both machine learning and statistical inference?

- ☐ Using population data to model a null hypothesis.
- ☐ Using population data to make inferences about a null sample.
- ☒ Using sample data to infer qualities of the underlying population distribution.
- ☐ Using sample data to make inferences about a hypothesis.

✔ Nice work

Correct. In both machine learning and statistical inference, we're using sample data to infer qualities of the underlying population distribution.

2. Which one of the following describes an approach to customer churn prediction stated in terms of probability?

- ☐ Data related to churn may include the target variable for whether a certain customer has left.
- ☐ Churn prediction is a data-generating process representing the actual joint distribution between our x and the y variable.
- ☒ Predicting a score for individuals that estimates the probability the customer will leave.
- ☐ Predicting a score for individuals that estimates the probability the customer will stay.

✔ Nice work

Correct. Churn prediction is often approached by predicting a score for individuals that estimates the probability the customer will leave.

3. What is customer lifetime value?

- ☒ The total purchases over the time which the person is a customer.
- ☐ The total value that the customer receives during their life.
- ☐ The total churn a customer generates in the population.
- ☐ The total churn generated by a customer over their lifetime.

4. Which one the following statements about the normalized histogram of a variable is true?

- ☐ It is a parametric representation of the population distribution.
- ☒ It provides an estimate of the variable's probability distribution.
- ☐ It serves as a bar chart for the null hypothesis.
- ☐ It is a non-parametric representation of the population variance.

✔ Nice work

Correct. The normalized histogram of a variable estimates the variable's probability distribution, and the estimate improves with the amount of data used.

5. The outcome of rolling a fair die can be modelled as a _____ distribution.

- ☐ Poisson
- ☐ log-normal
- ☒ uniform
- ☐ normal

✔ Nice work

Correct. The chance of rolling any particular value for a fair die is equally likely, so the outcome is uniformly distributed.

6. Which one of the following features best distinguishes the Bayesian approach to statistics from the Frequentist approach?

- ☐ Frequentist statistics requires construction of a prior distribution.
- ☐ Frequentist statistics incorporates the probability of the hypothesis being true.
- ☐ Bayesian statistics is better than Frequentist.
- ☒ Bayesian statistics incorporate the probability of the hypothesis being true.

7. Which of the following best describes what a hypothesis is?

- ☐ A hypothesis is a statement about a posterior distribution.
- ☐ A hypothesis is a statement about a sample of the population.
- ☒ A hypothesis is a statement about a population.
- ☐ A hypothesis is a statement about a prior distribution.

✔ Nice work

Correct. A hypothesis could be suggested by a sample of the population, but it is a statement about the entire population.

8. A Type 2 error in hypothesis testing is _____:

- ☐ correctly rejecting the alternative hypothesis.
- ☒ incorrectly accepting the null hypothesis.
- ☐ incorrectly accepting the alternative hypothesis.
- ☐ correctly rejecting the null hypothesis.

✔ Nice work

Correct. A type 2 error is incorrectly accepting the null hypothesis.

9. Which statement best describes a consequence of a type II error in the context of a churn prediction example? Assume that the null hypothesis is that customer churn is due to chance, and that the alternative hypothesis is that customers enrolled for greater than two years will not churn over the next year.

- ☐ You correctly conclude that customer churn is by chance
- ☐ You incorrectly conclude that there is no effect
- ☐ You correctly conclude that a customer will eventually churn
- ☒ You incorrectly conclude that customer churn is by chance

10. Which of the following is a statistic used for hypothesis testing?

- ☒ The likelihood ratio.
- ☐ The acceptance region.
- ☐ The standard deviation.
- ☐ The rejection region.

✔ Nice work

Correct. The likelihood ratio can be used as a test statistic, to decide whether to accept or reject the null hypothesis.